

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2020

Bachelor in Computer Application
Course Title: Operating System
Code No: CACS251
Semester: IV

Full Marks: 60
Pass Marks: 24
Time: 3 hrs

Candidates are required to answer all the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5=30]

2. What is an Operating System? Why is it known as the resource manager and the extended machine? [2+3]
3. What is Kernel? What role does it play in an Operating System? Differentiate between monolithic and micro kernel. [1+1+3]
4. Suppose that a disk has 200 cylinders (0-200). The head pointer is currently at 50 and previously at 35. The queue of pending request, in FIFO order is 60, 190, 40, 125, 15, and 150. The time required to move per cylinder is 6 msec. starting from the current head position, what the total seek time needed to satisfy all the pending request for each of the following disk scheduling algorithms? [2.5+2.5]
 - a) FCFS
 - b) SCAN
5. Define file and directory. Explain the concept of Access Control List (ACL) and Access Control Matrix (ACM). [2+3]
6. What is biometric password in authentication? Explain the various system threats. [2+3]
7. How distributed operating system is more applicable than centralized operating system? Explain the major goals of distributed operating system. [2+3]
8. Write short notes on any TWO: [2.5+2.5]
 - a) Producer Consumer problem
 - b) Coalescing and Compaction
 - c) Ubuntu

Group C

Attempt any TWO questions.

[2×10=20]

9. What is CPU scheduling? Write down the criteria for CPU scheduling? Consider the following set of processes, with the length of the CPU burst given in milliseconds, draw Gantt chart illustrating their execution and calculate average waiting time and turnaround time using:

- a) First Come First Serve
- b) Shortest Job First
- c) Non-preemptive priority (smaller number implies higher priority)
- d) Round Robin (quantum=1)

process	Burst Time	Priority
P0	10	3
P1	1	1
P2	2	5
P3	1	4
P4	5	2

The processes have arrived in the order P0, P1, P2, P3, P4 all time 0. [1+1+2+2+2+2]

10. Define the terms: page fault and thrashing. Consider the following page reference string: 1, 3, 5, 1, 7, 1, 5, 5, 1, 4, 3, 7, 6, 3, 4, 1. How many page faults would occur for each of the following page replacement algorithms assuming 3 page frames?

- a) FIFO page replacement
- b) LRU page replacement
- c) Optimal page replacement

[2+2+2+2+2]

11. Define deadlock. List out the conditions that can result in resource deadlock. Explain different deadlock handling methods in details. [2+1+7]